

Detectors of Nuclear Radiation:

The emission of charged particles such as α -particles, β -particles, protons as well as radiation like γ -rays take place in most of the nuclear reactions. In order to understand these particles and their interaction with atomic nuclei, precise information about their mass, momentum, energy etc. are necessary. Several nuclear radiation detectors depend for their operation on the ionization that is produced in them by the passage of charged particles. Some of the detectors are, ionization chambers, G-M counters, cloud chambers etc. The phenomenon of excitation and molecular dissociation are also employed in the detection of radiation.

The Ionization Chamber:

The working principle employed in ionization chamber is that charged subatomic particles when pass through the matter/gas can ionise the matter/gases. The number of ion-pairs produced gives us information not only on the nature of incident particles but also about their energy.

Construction:

The ionisation chamber consists of a hollow metallic cylinder C , closed at both ends. There is a window W at one of the ends for the entry of the ionizing particles or radiations. A metallic rod R , well insulated from the cylinder, is mounted coaxially within the cylinder. The rod is connected to a quadrant electrometer E . A potential difference of ($\sim 10^2$ V) is maintained between C and R . An earthed guard ring G prevents leakage of charge from the cylinder to the rod. The chamber contains some gas like SO_2 , CH_3Br .

Working:

When charged particles enter the chamber it collides with the matter particles. As a result a large number of ion pairs are produced along its path. The positive ion is attracted towards the rod R and negative ion towards the wall of the chamber C . The quadrant electrometer E measures the rate of deposition of positive charge on R .

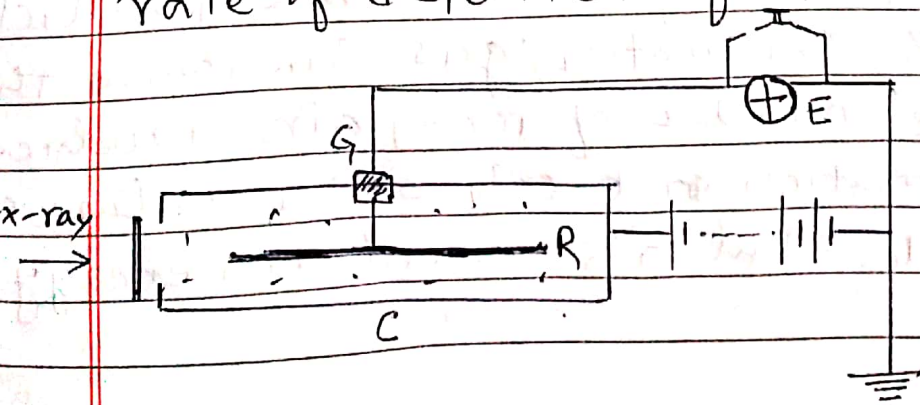


fig. Ionization chamber
i.e. detector of radiation

Due to the flow of ionized charge a feeble current is produced which is in the order of 10^{-12} to 10^{-15} amperes. Special electrometers and D.C. amplifying devices have to be employed to measure such small currents. The individual particles can also be counted using a pulse amplifier.

Uses:

The ionization chambers have been used to study α -particles, β -particles, protons, electrons & nuclei of lighter elements. It can also be used for measurements on X-rays & γ -ray. It can also be used for detection of chargeless particle like neutron; for which the chamber is filled with Boron Trifluoride vapour.

Geiger-Muller Counter (G-M Counter): -

A Geiger Counter (Geiger-Muller tube) is a device used for the detection & measurement of all types of radiation: Such as α - and β -particle and γ -radiation.

Construction: It consists of a metal chamber containing air or some other gases at a pressure of about 70 cm of Hg. A fine tungsten wire (W) is stretched along the

axis of the tube and is insulated from it by ebonite plugs E, E. The wire is connected to the positive terminal of a high tension supply ($\sim 1000\text{V}$ to 3000Volts) through a high resistance R ($\sim 100\text{ M}\Omega$) and the negative terminal is connected to the chamber C. The d.c. voltage is kept slightly less than that which will cause a discharge between the electrodes.

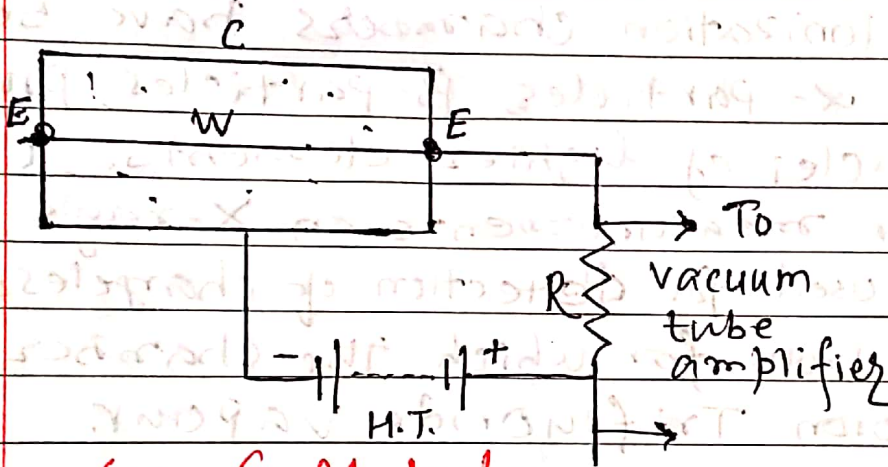


fig. G-M tube.

Normally, the length of G.M. tube is $10-100\text{ cm}$ and diameter $0.3\text{ cm}-10\text{ cm}$. The tungsten wire with thickness $2-5 \times 10^{-3}\text{ cm}$.

The press inside the tube is $5-10\text{ cm Hg}$.

The gas may be Br-vapour, Alcohol-vapour, Ar, SO_2 , CH_3Br etc.

Working:

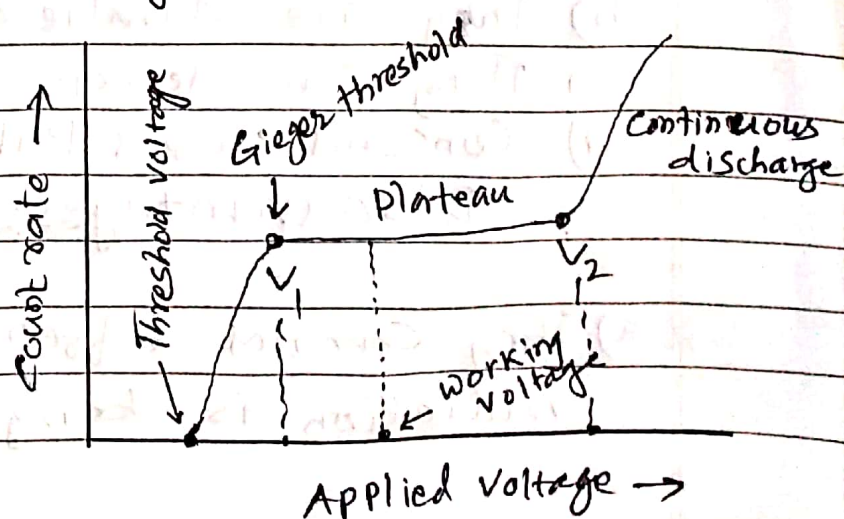
When ionizing radiation such as an alpha, beta or gamma particle enters the tube, it can ionize some of the gas molecules in the tube. From these ionized atoms, an electron is knocked out of the atom, and the remaining is positively charged. The high voltage in the tube produces an electric field inside the tube. The knocked out electrons are attracted to the +ve electrode and the positive ions are attracted towards -ve electrodes.

This produces a pulse of current in the wires connecting the electrodes and hence the pulse is counted. After the pulse is counted, the charged ion becomes neutralized and the counter becomes ready to record another pulse. A gas added to the tube helps to restore quickly to its original state. In this way single particle can also be registered.

For low voltage, electric field is much lower to record the pulses. The voltage at which G-M tube just begins to count is called the starting potential or threshold potential. The count rate rapidly rises and forms a 'plateau' region. The range of voltage is called 'plateau region'. The potential at which plateau begins is called Geiger threshold. The plateau region also is known as Geiger region. Eventually the voltage becomes too high and ~~etc~~ occurs continuous discharge, which is unwanted for the experiment. For the best operation, the voltage should be selected close to the threshold voltage and within the first $\frac{1}{4}$ th of the way into the plateau region. In the plateau region, the counting rate is, more or less, independent of small changes in p.d. across the tube.

Efficiency:

The counting efficiency is defined as the ability of its counting, if at least one ion pair is produced in it.



The efficiency of a detector is given by

$$\eta = \frac{\text{number of particles of radiation detected}}{\text{number of particles of radiation emitted}}$$

The counting efficiency can also be expressed as

$$\eta = 1 - e^{-slp}$$

where s = specific ionization at one atmosphere.

p = pressure in atmosphere

l = path length of ionization particle in the counter.

Resolving Time (Dead Time):

After a count has been recorded, it takes certain amount of time to reset itself to be ready to record the next count. The dead time is the time taken by the tube to "reset" itself. Since the detector is "not operating" while it is being reset, the measured activity is not the true activity of the sample.

Advantages

- They are relatively inexpensive.
- They are durable and easily portable.
- They can detect all kinds of radiations
- Constant output Pulse size, independent of initial ionization.

Disadvantages:

- They cannot differentiate which type of radiation is being detected.

- b) They can not be used to determine the exact energy of the detected radiation.
c) They have a very low efficiency.

Relative plateau slope:

The slope of the plateau is the average percentage change in the counting rate per unit increment of applied voltage and is known as relative plateau slope (R.P.S).

$$R.P.S = \left(\frac{N_2 - N_1}{N} \right) \times 100 \div (V_2 - V_1)$$

where, N_1 = Counting rate at potential V_1
 N_2 = " " " " " V_2

$$N = \frac{N_1 + N_2}{2}$$

operating voltage

$$V = V_1 + \frac{1}{3} (V_2 - V_1) \text{ Volts.}$$

*

Further,

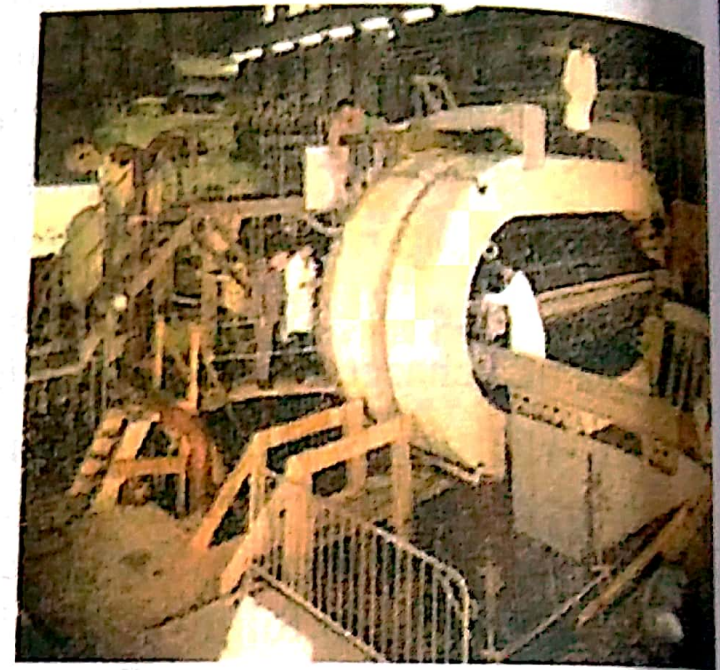
- i) Bubble chamber
- ii) The Scintillation counter
- iii) Cerenkov counter
- iv) Bubble chamber
- v) Semiconductor detector

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29.9 Bubble Chamber ✓

Principle. We know that normally the liquid boils with the evolution of bubbles of vapour at the boiling point. If the liquid is heated under a high pressure to a temperature well above its normal boiling point, a sudden release of pressure will leave the liquid in a *superheated state*. If an ionising particle passes through the liquid within a few milliseconds after the pressure is released, the ions left in the track of a particle act as condensation centres for the formation of vapour bubbles. The vapour bubbles grow at a rapid rate and attain a visible size in a time of the order of 10 to 100 μs . Thus in a bubble chamber, a vapour bubble forms in a superheated liquid, whereas in a cloud chamber, a liquid drop forms in a supersaturated vapour. Thus an ionising particle passing through the superheated liquid leaves in its wake a trail of bubbles which can be photographed.

A schematic diagram of a liquid hydrogen bubble chamber, operating at a temperature of 27 K is shown in Fig. 29.13. A box of thick glass walls is filled with liquid hydrogen and connected to the expansion pressure system. To maintain the chamber at constant temperature, it is surrounded by liquid nitrogen and liquid hydrogen shields. High energy particles are allowed to enter the chamber from the side window *W*. A sudden release of pressure from the expansion valve is followed by light flash and camera takes the stereoscopic view of the chamber.



Bubble Chamber.

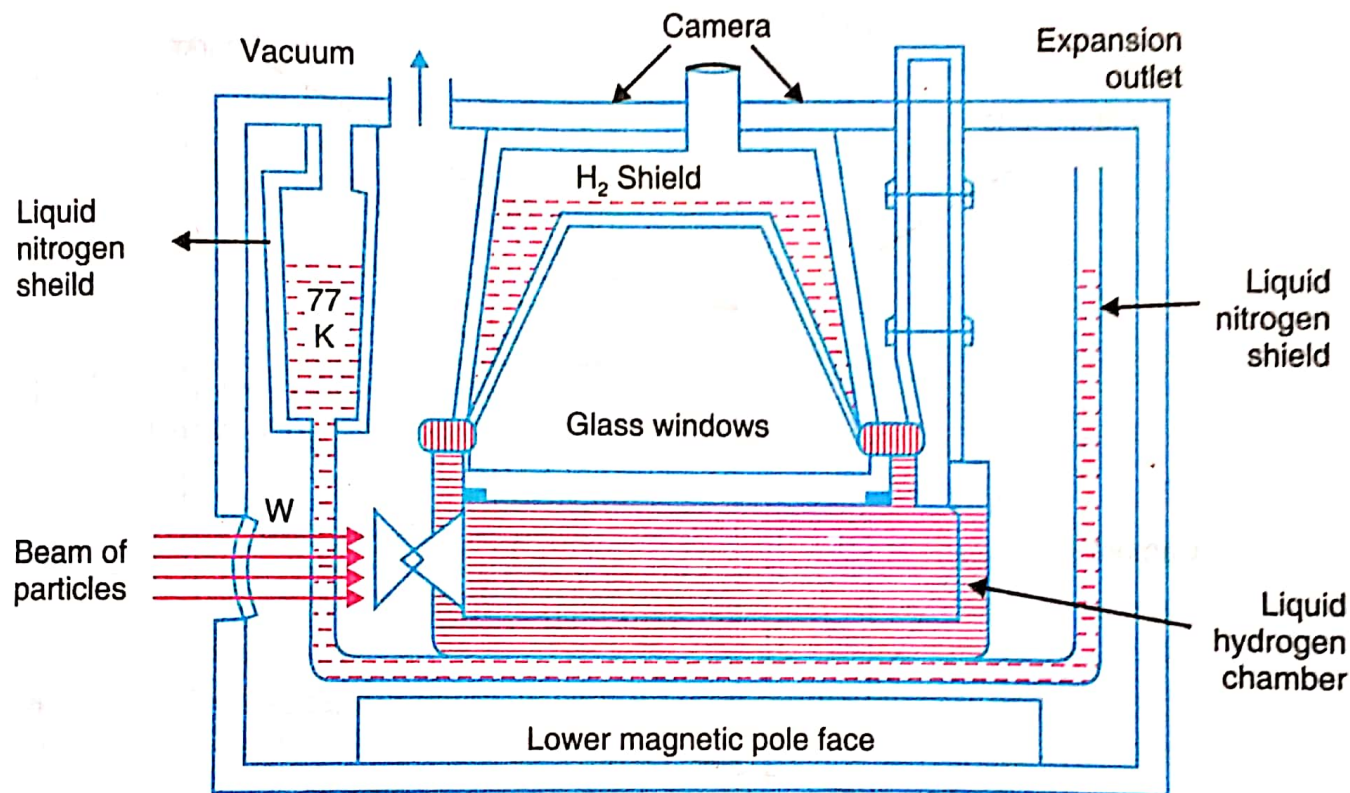


Fig. 29.13

The incoming beam triggers the chamber. The charge of the tracks can be identified by the direction of their curvature in the magnetic field applied over the bubble chamber. From the curvature and length of the track, the momentum and energy of the particle can be found. The bubble chamber is used to *study particle interaction* and to *detect very high energy particles*.

Advantages. (1) The density of a liquid is very large when compared to that of a gas of even high pressure. Hence the chances of collision of a high energy particle with a molecule of the liquid are very much greater. Consequently there is a greater chance of their track being recorded. So the chances of recording events like cosmic ray phenomena are improved when compared with cloud chambers.

(2) The bubbles grow rapidly and as a result the tracks are not likely to get distorted due to convection currents in the liquid.

(3) The bubble chamber is sensitive even to particles of low ionising power.

29.12 The Scintillation Counters ✓

One of the earliest methods of radiation detection was the spinthariscopes (Fig. 29.15). It consists of a small wire, the tip of which is dipped in Radium bromide (R) or any other radioactive salt. It is placed in front of a zinc



Scintillation Counter.

sulphide screen S and viewed through a microscope.

When an α or β -particle falls on the zinc sulphide screen, they produce light flashes which can be seen by a microscope (M) in a dark room. The visible luminescence excited in zinc sulphide by α -particles was used by Rutherford for counting the particles. The process of counting these scintillations through

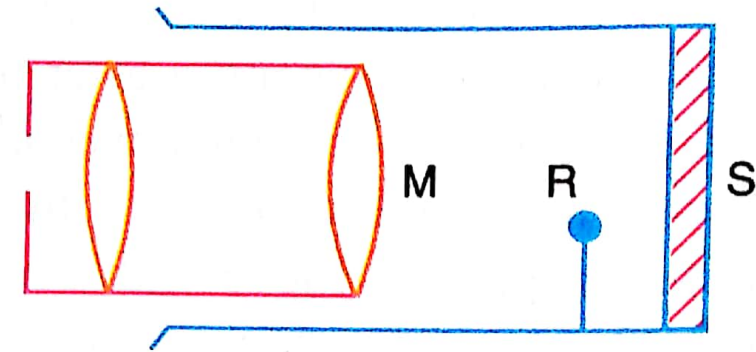


Fig. 29.15

a low power microscope is a tedious one and the limitations of observation with the eye restrict the counting rate to about 100 per minute. This process, whereby the energy of the particle is converted to light, is the basis of scintillation counter.

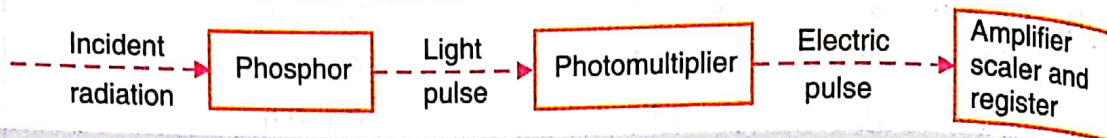


Fig. 29.16

The main parts of a scintillation counter are shown in Fig. 29.16. The atoms of the phosphor are excited or ionised by the energy loss of an impinging α , β or γ ray. When the atoms return to their ground states, photons are emitted, in the blue and ultraviolet regions of the optical spectrum. The phosphor is optically coupled to the envelope of a photomultiplier tube. The photons strike the photocathode, causing the ejection of photo-electrons (Fig. 29.17). As these photo-electrons leave the photocathode, they are directed by a focusing electrode to the first multiplier electrode or *dynode*. This electrode has the property of emitting three, four or five electrons for every single

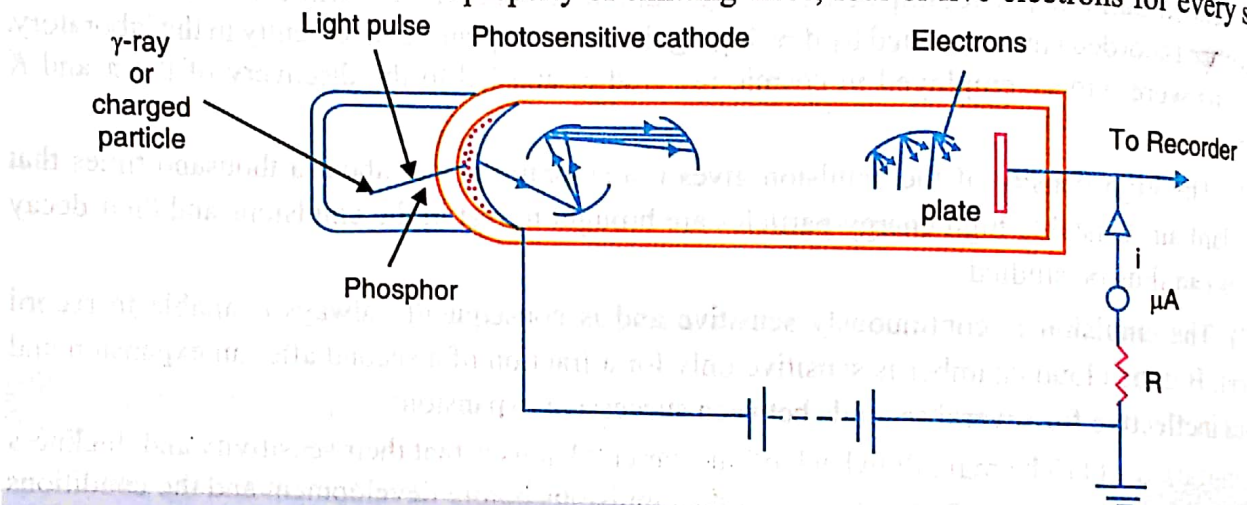


Fig. 29.17

electron which strikes its surface. There may be from 10 to 14 such multiplier stages in a given tube. Hence, from the emission of one single electron from the cathode, a burst of one million electrons may impinge on the final stage in the tube (the anode). The output pulse from the photomultiplier is fed to a pulse amplifier followed by a scaler circuit.

29.13 Cerenkov Counter ✓

When a charged particle moves through a transparent dielectric medium with a velocity greater than the velocity of light in that medium, a cone of light waves is emitted. These light waves are known as *Cerenkov radiation*. In a dielectric medium of refractive index n , photons move with a velocity c/n . Consider a charged particle moving with a velocity v through the dielectric medium. Fig. 29.18 shows the interactions when $v > c/n$. The envelope of the radiation is a cone of half angle θ with the particle at its apex, where

$$\sin \theta = \frac{(c/n)t}{vt} = \frac{c}{nv}$$

The angle θ of the cone of radiation depends on the speed v of the particle. When a beam of fast charged particles moves through a medium such as glass or transparent plastic, the radiation can be detected and the angle θ measured. Thus the speed of the particle can be determined.



Cerenkov Counter.

Fig. 29.19 shows a Cerenkov detector designed by Marshall. A collimated beam of pions is allowed to pass along the axis of a large hemispherical perspex lens. The Cerenkov radiation radiated in this lens is reflected by the cylindrical mirror M_1 on to one of the plane mirrors M_2 , M_3 and then on to the cathodes of the photomultiplier tubes (IP 28). The two photomultipliers operate in coincidence. The position of the radiator lens is adjusted so that the Cerenkov light cone is correctly focused on to the phototubes to give maximum coincidence counting rate.

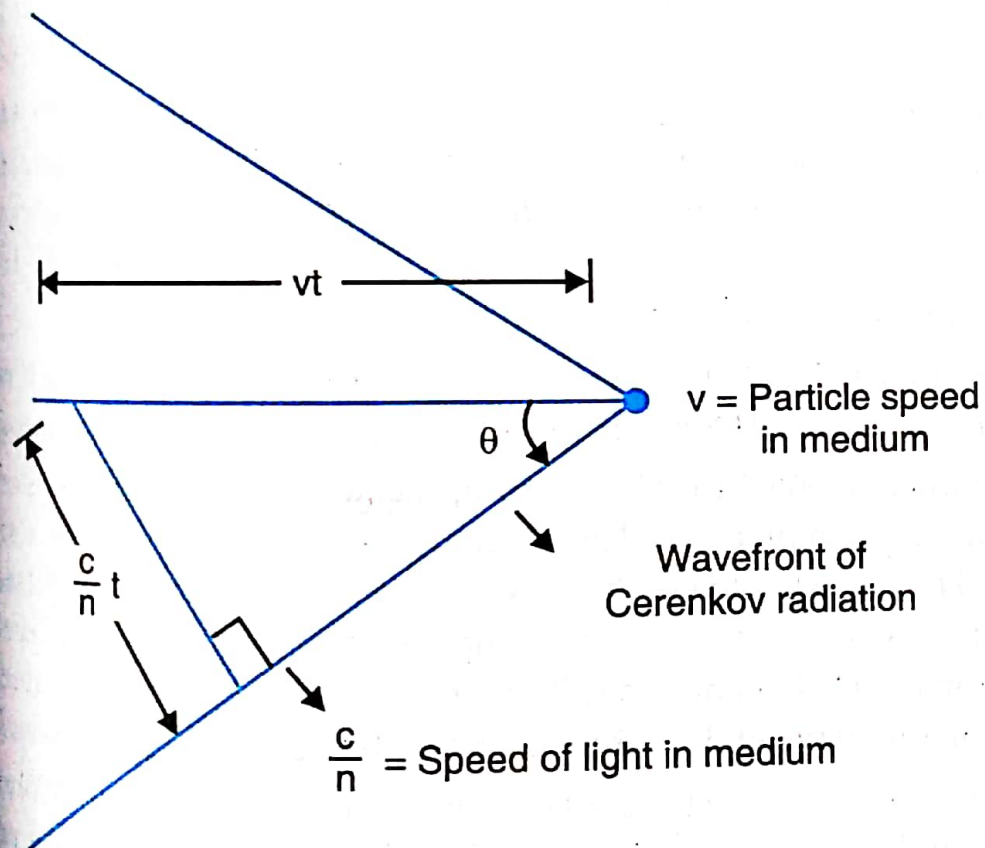


Fig. 29.18

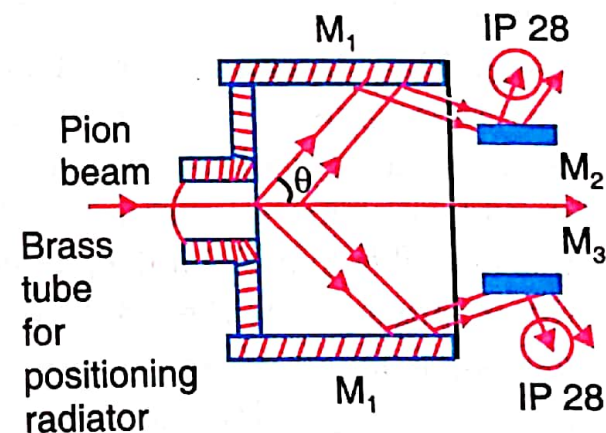


Fig. 29.19

The Cerenkov detector is used for the detection of high energy particles. It has a high efficiency and a high counting rate.